

Polynomial Queries

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to maintain an array of n values and efficiently process the following types of queries:

1. Increase the first value in range $[a, b]$ by 1, the second value by 2, the third value by 3, and so on.
2. Calculate the sum of values in range $[a, b]$.

Input

The first input line has two integers n and q : the size of the array and the number of queries.

The next line has n values $t_1, t_2, \dots, t_n$: the initial contents of the array.

Finally, there are q lines describing the queries. The format of each line is either "1 a b " or "2 a b ".

Output

Print the answer to each sum query.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq t_i \leq 10^6$
- $1 \leq a \leq b \leq n$

Example

Input	Output
5 3 4 2 3 1 7 2 1 5 1 1 5 2 1 5	17 32